

An Automated Question Generation (AQG) Model based on Natural Language Processing

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Dissertation Approval Sheet

The dissertation entitled “An Automated Question Generation (AQG) Model using Natural Language Processing” submitted by Anish Gupta, Divyansh Dua and Ishu Anand Malviya is approved as partial fulfillment for the award of Bachelor of Engineering in Computer Engineering degree by Devi Ahilya Vishwavidyalaya, Indore.

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Recommendation

The dissertation entitled "**An Automated Question Generation (AQG) Model based on Natural Language Processing** " Submitted by Anish Gupta, Divyansh Dua and Ishu Anand Malviya is a satisfactory account of the Bonafide work done under my supervision and is recommended towards the partial fulfilment for the award of Bachelor of Engineering in Computer Engineering degree by Devi Ahilya Vishwavidyalaya, Indore.

Date: 7th November 2025

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Candidate Declaration

We hereby declare that the work which is being presented in this project entitled "**An Automated Question Generation (AQG) Model based on Natural Language Processing**" in partial fulfillment of degree of **Bachelor of Engineering in Computer Engineering** is an authentic record of our own work carried out under the supervision and guidance of **Mr. Arpit Agrawal**, Faculty in Department of **Computer Engineering**, Institute of Engineering and Technology, Devi Ahilya Vishwavidyalaya, Indore

We are fully responsible for the matter embodied in this project in case of any discrepancy found in the project and the project has not been submitted for the award of any other degree.

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ABSTRACT

This project presents an Automated Question Generation (AQG) system developed using the T5 (Text-to-Text Transfer Transformer) model and Natural Language Processing (NLP) techniques. The objective of the system is to automatically generate meaningful, contextually relevant, and grammatically correct questions from a given text passage. The project addresses the limitations of manual question preparation, which is often time-consuming, repetitive, and dependent on subject expertise. By automating this process, the system aims to support educators, students, online learning platforms, and assessment systems.

The proposed model was fine-tuned using a filtered subset of the Stanford Question Answering Dataset (SQuAD), containing high-quality context-question pairs. The implementation was carried out using the Hugging Face Transformers library, PyTorch, and Google Colab with GPU acceleration. During preprocessing, the dataset was tokenized and converted into a text-to-text format suitable for the T5 model. The model was trained using the AdamW optimizer and evaluated using BLEU and ROUGE metrics to measure the quality and fluency of generated questions.

The developed system is capable of generating multiple descriptive questions from arbitrary input text while maintaining semantic relevance and linguistic coherence. To improve usability, a Graphical User Interface (GUI) was implemented using Gradio, enabling users to interact with the model through a web-based interface. Additional functionalities such as batch question generation, REST API integration, and optimized decoding using beam search were also incorporated.

Experimental results demonstrate that the fine-tuned T5 model effectively generates high-quality questions with strong contextual understanding and natural language fluency. The project highlights the practical applicability of transformer-based NLP models in educational technology, automated assessments, intelligent tutoring systems, and content generation platforms.

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1.1 Overview and Issues Involved

Question generation is an important task in the field of Natural Language Processing (NLP) and educational technology because it helps in creating assessments, quizzes, and practice material automatically from textual content. In traditional educational systems, teachers manually prepare questions from books, notes, and study material. This process requires considerable time, effort, and subject expertise. As educational content on digital platforms continues to increase rapidly, manually generating large numbers of questions becomes difficult and inefficient.

The development of transformer-based language models has significantly improved automatic text generation tasks. Models such as T5 (Text-to-Text Transfer Transformer) are capable of understanding the contextual meaning of a text passage and generating human-like textual outputs [1]. In this project, the T5 transformer model is fine-tuned for automated question generation using the SQuAD dataset [2]. The model accepts a text passage as input and generates relevant questions based on the context provided.

The implemented system is primarily designed to generate factual and context-based questions rather than long descriptive or analytical questions. Since the model is trained on the Stanford Question Answering Dataset (SQuAD), it learns patterns from short question-answer pairs created from Wikipedia passages [2]. As a result, the generated questions are generally concise and focused on extracting important factual information from the input text. The system mainly produces questions beginning with terms such as “What,” “Who,” “When,” “Where,” and “How,” depending on the context available in the passage.

Although automated question generation systems have improved significantly, several challenges still exist. Many traditional systems depend on handcrafted grammatical rules and templates, which limit flexibility and adaptability across different domains. Earlier sequence-to-sequence models also faced difficulty in maintaining contextual understanding for long passages and often generated grammatically incorrect or irrelevant questions [3]. Even transformer-based models can face issues such as repetitive outputs, limited creativity, and dependence on the quality of training data.

The proposed project focuses on developing a transformer-based Automated Question Generation (AQG) system using the T5-small model and modern NLP techniques. The system performs

preprocessing, tokenization, contextual encoding, and question generation through an encoder–decoder transformer architecture. To improve accessibility and usability, a web-based Graphical User Interface (GUI) was developed using Gradio, allowing users to generate questions interactively from any input passage.

Therefore, there is a need for an efficient transformer-based system capable of automatically generating meaningful factual questions directly from textual content while maintaining contextual relevance and grammatical correctness. The proposed project attempts to address these challenges using the T5-small transformer model.

1.2 Statement of the Problem

The rapid growth of digital learning platforms, online education systems, and electronic study material has increased the demand for automatic assessment and question generation systems. In traditional educational environments, teachers and educators manually prepare questions from textbooks, lecture notes, and study resources. This process requires significant time, effort, and subject expertise. Generating large numbers of meaningful questions manually becomes difficult, especially when educational content continuously increases across multiple domains.

Most existing question generation systems rely on handcrafted grammatical rules, predefined templates, or complex linguistic processing methods. These systems often lack flexibility and perform poorly when handling different writing styles or complex textual passages. Earlier sequence-to-sequence deep learning models also faced limitations in understanding long contextual relationships and frequently generated repetitive, irrelevant, or grammatically incorrect questions.

The major problem addressed in this project is the development of an automated system capable of generating contextually relevant and grammatically meaningful questions directly from raw textual input using Natural Language Processing techniques. The system should minimize manual intervention and generate questions automatically without depending on manually designed rules or external linguistic resources.

The project specifically focuses on generating factual and context-based questions rather than long descriptive or analytical questions. Since the T5-small model is trained on the SQuAD dataset, the generated outputs are expected to follow question patterns learned from context–question pairs present in the dataset. The challenge is to ensure that the generated questions remain semantically relevant to the input passage while maintaining grammatical correctness and contextual

understanding.

To address these problems, the project proposes a transformer-based Automated Question Generation (AQG) system using the T5 model. The system aims to automate question generation efficiently and provide a practical solution that can be used in educational platforms, online examinations, intelligent tutoring systems, and assessment applications.

1.3 Significance of Problem

Automatic Question Generation (AQG) has become an important research area in Natural Language Processing and educational technology because of the growing demand for intelligent learning systems and automated assessments. Educational institutions, online learning platforms, coaching systems, and examination portals require large numbers of questions for assignments, quizzes, practice tests, and evaluations. Preparing these questions manually is a repetitive and time-consuming task that requires subject expertise and continuous effort from educators.

Historically, early question generation systems were based mainly on handcrafted rules, syntactic parsing, and predefined templates [3]. Although these approaches could generate simple questions, they lacked flexibility and could not adapt effectively to different domains or writing styles. Later, sequence-to-sequence deep learning models improved the fluency of generated text but still faced difficulties in maintaining long-range contextual understanding and semantic consistency.

The rapid advancement of transformer architectures has significantly improved natural language understanding and generation tasks in recent years. Models such as T5 have demonstrated strong performance in text generation applications because of their ability to capture long-range contextual dependencies using self-attention mechanisms [1]. This advancement makes transformer-based AQG systems more effective than earlier rule-based and sequential approaches.

The significance of this project lies in developing an automated question generation system capable of generating meaningful factual questions directly from textual passages using a transformer-based approach. The system reduces the manual effort required for question preparation and provides a scalable solution for generating assessment content automatically. It can help teachers create practice material more efficiently and support students by providing dynamically generated questions for learning and revision purposes.

This project also demonstrates the practical application of modern NLP techniques in educational

systems. By fine-tuning the T5-small model on the SQuAD dataset and integrating it with a user-friendly Gradio interface, the system provides an efficient and accessible platform for automatic question generation. The developed model can be applied in online examination systems, intelligent tutoring platforms, adaptive learning environments, and educational content generation systems where automatic creation of context-based questions is required.

1.4 Assumptions

Certain assumptions were considered during the development and implementation of the Automated Question Generation (AQG) system to ensure proper functionality and reliable output generation. These assumptions help define the scope and limitations of the proposed system.

The first assumption is that the input text provided by the user is grammatically correct, meaningful, and written in proper English language. Since the quality of generated questions depends directly on the quality of the input passage, incomplete or poorly structured text may reduce the relevance and correctness of generated questions.

The project assumes that the SQuAD dataset used for training and fine-tuning contains accurate and well-structured context-question pairs. The T5-small model learns question generation patterns from this dataset; therefore, the effectiveness of the generated output depends on the quality and diversity of the training data.

Another assumption is that the T5 transformer model has sufficient capability to understand contextual relationships within textual passages and generate semantically relevant questions without requiring handcrafted grammatical rules or predefined templates. The system relies entirely on transformer-based learning methods rather than traditional rule-based approaches.

The project also assumes the availability of adequate computational resources such as GPU-enabled cloud environments provided by Google Colab. Transformer-based models require high computational power and memory for training, fine-tuning, and inference operations.

It is further assumed that the generated questions will mainly be factual and context-based rather than highly analytical or long descriptive questions. Since the model is trained using SQuAD data, the generated outputs are expected to follow similar question-generation patterns learned from the dataset.

Finally, the system assumes that users will provide textual input only and not images, handwritten content, or noisy data. The project is specifically designed for processing English text passages and generating questions automatically from textual content.

1.5 Proposed Solution

The proposed solution for this project is the development of an Automated Question Generation (AQG) system using the T5 (Text-to-Text Transfer Transformer) model and Natural Language Processing (NLP) techniques. The system is designed to automatically generate meaningful, grammatically correct, and contextually relevant questions from a given text passage without relying on manually created grammatical rules or predefined templates.

The implemented system follows a transformer-based encoder–decoder architecture. In this approach, the input text passage is first preprocessed and tokenized into a format suitable for the T5 model. The encoder processes the contextual information present in the text and converts it into contextual embeddings, while the decoder generates questions based on the learned semantic relationships within the passage.

The project uses the Stanford Question Answering Dataset (SQuAD) for fine-tuning the T5-small model [2]. The dataset contains high-quality context–question pairs that help the model learn how factual and context-based questions are formed from textual passages. During training, the model learns patterns between passages and corresponding questions so that it can generate relevant questions for unseen input text.

The developed system provides several functionalities including automatic question generation from input text passages, multiple question generation, contextual understanding using transformer architecture, web-based interaction through Gradio GUI, batch input processing, and efficient question decoding using beam search. These functionalities help improve usability and make the system suitable for educational and assessment applications.

To improve accessibility and usability, a Graphical User Interface (GUI) was developed using Gradio. The interface allows users to enter any text passage and instantly generate questions through an interactive web-based application. The system also supports multiple question generation and batch processing functionality for handling larger text inputs efficiently.

The workflow of the proposed system can be represented as follows:

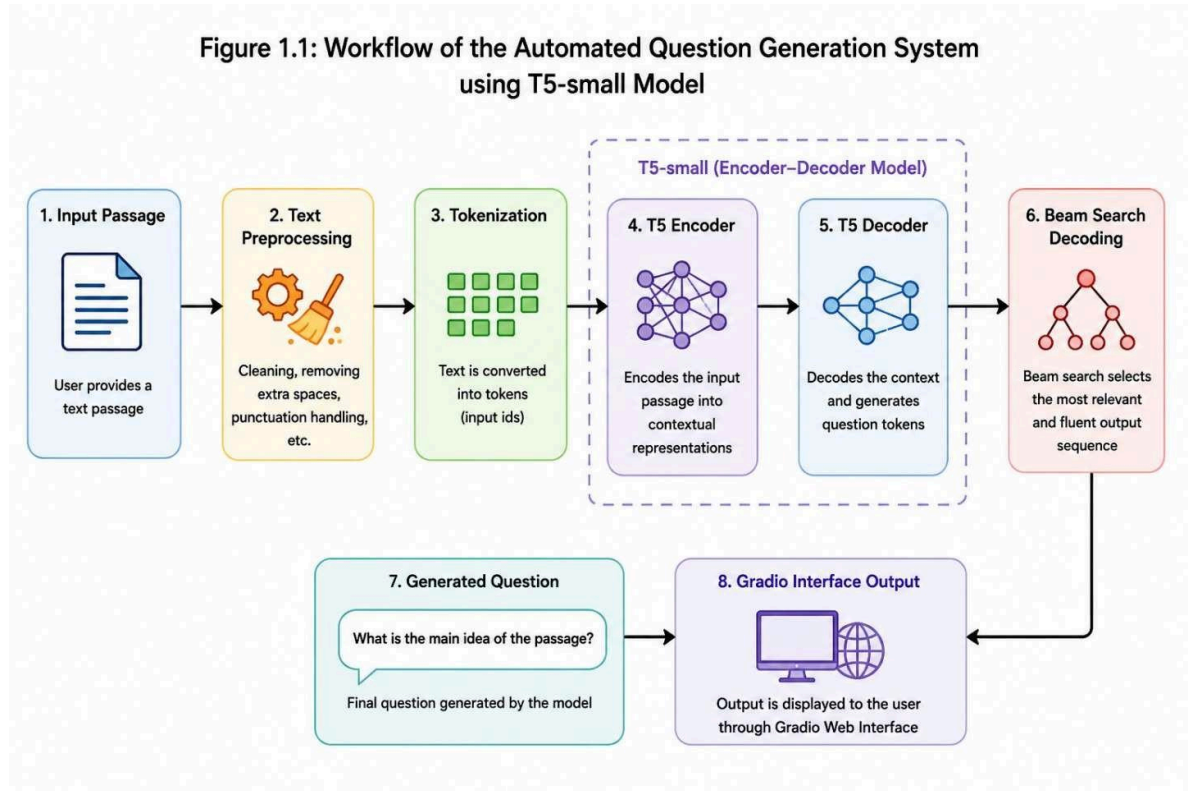


Figure 1.1 illustrates the workflow of the proposed Automated Question Generation system using the T5-small transformer model.

The generated outputs are primarily factual and context-oriented questions such as “What,” “Who,” “When,” “Where,” and “How” type questions. To improve the quality and fluency of generated questions, beam search decoding is used during inference [4]. This helps the model select the most meaningful and grammatically correct output sequence from multiple possible predictions.

Compared to traditional rule-based and template-based question generation systems, the proposed transformer-based approach provides better contextual understanding, improved grammatical fluency, and higher flexibility across different text domains. Unlike earlier systems that depended heavily on handcrafted linguistic rules, the T5-small model learns semantic relationships directly from data, making the system more scalable and adaptable for real-world educational applications.

2.1 Existing Papers

Addressing the high cost and scarcity of large-scale, human-labeled datasets for Question Answering, Fabbri et al. [1] proposed a template-based question generation framework for unsupervised QA that retrieves semantically related sentences and applies structured templates to form questions. This approach avoids the need for manually created context–question pairs. Instead, the system identifies an answer in a context, retrieves a semantically similar sentence containing the same answer, and uses that sentence as the basis for question generation. Their method improved F1 scores by 14–20% over previous unsupervised baselines while using fewer synthetic samples. The major advantage of this approach is reduced dependence on labeled datasets and improved diversity in generated training examples. However, limitations remain in the grammatical fluency of generated questions because template-based approaches cannot always guarantee natural sentence structures. The model also faced difficulty in handling non-entity answers and required further testing for multilingual applications.

Rahim et al. [2] presented an automated examination question generator using a Genetic Algorithm (GA) to generate multiple-choice questions aligned with Bloom’s Taxonomy. The system categorized questions into different cognitive levels and optimized examination papers according to educator-defined constraints such as topic coverage and difficulty level. Experimental results showed an average question quality of around 70%, demonstrating the effectiveness of the optimization approach. The GA-based system improved diversity and reduced repetitive question patterns. However, the model depended completely on an existing question bank and could not generate entirely new questions dynamically from raw textual passages.

Chang and Chung [3] developed a hybrid question classification system that combined machine learning techniques with keyword-based mapping to Bloom’s Taxonomy. Their framework classified questions into different cognitive levels such as remembering, understanding, and analysis. The system achieved approximately 75% accuracy for lower-level cognitive classification. Although the keyword-based approach worked effectively for simpler questions, it struggled to capture the semantic complexity of higher-order analytical questions because keyword matching alone could not fully understand contextual meaning.

Chan and Fan [4] introduced a BERT-based approach for automatic question generation to

overcome the limitations of traditional recurrent neural network models in handling long contextual passages. Their model used transformer-based self-attention mechanisms to capture global contextual dependencies more effectively. The proposed BERT-HLSQG model inserted special highlight tokens around answer spans to improve contextual focus during question generation. Tested on the SQuAD dataset, the model achieved significant improvement in BLEU-4 score from 16.85 to 22.17 and generated more fluent and human-like questions. However, the approach required high computational resources and extensive training time for efficient performance.

Atharva et al. [5] developed a T5-based automatic question generation system capable of generating both multiple-choice and subjective questions for educational assessments. Their system fine-tuned the T5 transformer model as the main generative engine and also incorporated distractor generation for MCQs. An additional evaluator model was used to filter low-quality generated outputs and improve overall question quality. The model achieved 76% syntactic accuracy and 84% fluency for generated question-answer pairs. This work demonstrated the effectiveness of transformer-based architectures in generating structured and contextually meaningful educational content.

From the reviewed literature, it is observed that transformer-based architectures provide better contextual understanding and question generation capabilities compared to traditional rule-based and sequential models. Most modern systems use pretrained language models such as BERT and T5 for generating fluent and contextually relevant questions. Based on these observations, the proposed project uses the T5-small transformer model fine-tuned on the SQuAD dataset to generate factual and context-based questions automatically from textual passages.

2.2 Technologies and Research Tools

2.2.1 Google Colab

Google Colab was used as the primary development environment for training and fine-tuning the T5 model. It provides a cloud-based Jupyter Notebook interface along with free GPU support, which is essential for training transformer-based deep learning models. The availability of NVIDIA T4 GPUs in Google Colab enabled efficient model training within the limitations of a semester project.

2.2.2 Google Scholar

Google Scholar was used during the literature review and research phase to collect research papers related to Automated Question Generation (AQG), Natural Language Processing (NLP), transformer models, and the SQuAD dataset. It helped in understanding existing methodologies and

identifying research gaps relevant to the project.

2.2.3 Hugging Face Transformers

The Hugging Face Transformers library was used for loading the pretrained T5-small model and tokenizer. It provides APIs for transformer-based NLP tasks such as text generation, summarization, translation, and question generation. The T5-small model used in this project was loaded and fine-tuned using this library.

2.2.4 PyTorch

PyTorch was used as the deep learning framework for model training, fine-tuning, tensor operations, and optimization. It provides dynamic computation graphs and efficient GPU acceleration support required for transformer-based architectures.

2.2.5 SQuAD Dataset

The Stanford Question Answering Dataset (SQuAD) was used for training and evaluation. The dataset contains high-quality context-question pairs extracted from Wikipedia articles, making it suitable for question generation tasks. The model learns contextual relationships between passages and questions from this dataset.

2.2.6 NLTK

The Natural Language Toolkit (NLTK) was used for preprocessing operations such as tokenization and text handling. It provides utilities for processing textual data before feeding it into the transformer model.

2.2.7 Gradio

Gradio was used to create the web-based Graphical User Interface (GUI) for the project. The interface allows users to enter any text passage and generate questions interactively through a browser-based application.

2.2.8 Git

Git was used for version control and managing modifications made during project development. It helped track code changes, experimental updates, and different implementation versions efficiently.

2.2.9 GitHub

GitHub was used as the cloud-based repository platform for storing project source code, notebooks, and documentation securely. It also served as a backup and collaboration platform during development.

2.2.10 draw.io

draw.io was used to create architecture diagrams, system flowcharts, and workflow illustrations included in the project report. These diagrams help explain the implementation pipeline and subsystem interactions clearly.

3.1 Description of Research Design and Procedures Used

Our project's goal was to teach an AI to read a piece of text (a context) and write a good, open-ended question about it. To do this, we used a supervised learning approach, which is like teaching a student by showing them thousands of examples. We showed the AI a context and a human-written question, and it gradually learned the pattern. This research project aims to build an AI-powered Question Generation System using a fine-tuned T5 transformer model. The research follows a supervised learning method where the model learns patterns from context–question pairs. The entire workflow includes data preparation, model development, training, evaluation, and GUI creation.

3.1 Conceptual Requirements

3.1.1 Text-to-Text Transfer Transformer

The T5 (Text-to-Text Transfer Transformer) model is a powerful and highly flexible neural network architecture developed by Google to unify a wide range of Natural Language Processing tasks under a single text-to-text framework. Instead of treating tasks like translation, summarization, classification, or question answering as separate problems that require different input and output formats, T5 converts all tasks into a simple format where the input is text and the output is also text. This consistency makes the model easy to fine-tune, reuse, and deploy across diverse applications. T5 is built on the encoder–decoder transformer architecture, originally introduced in the “Attention is All You Need” paper, enabling it to understand input sequences through the encoder and generate meaningful output sequences through the decoder. One of the key innovations in T5 is its training on the massive C4 (Colossal Clean Crawled Corpus) dataset, which consists of cleaned web text and provides the model with a rich understanding of language patterns. The model is pretrained using an objective called span corruption, where random spans of text are masked and the model learns to predict them, improving its ability to generate coherent and contextually relevant outputs. T5 comes in several sizes, ranging from small models to extremely large versions with billions of parameters, allowing developers and researchers to choose variants depending on computational needs. Its versatility enables it to achieve high performance on tasks such as translation, reading comprehension, question generation, summarization, and text classification. By framing all tasks in a unified way, T5 has significantly simplified the development of NLP systems and pushed state-of-the-art results across multiple benchmarks.

3.1.2 Transformers Library

The **Transformers** library from Hugging Face is an open-source, industry-standard toolkit that centralizes state-of-the-art pretrained transformer models (like BERT, GPT, RoBERTa, T5, DistilBERT, and many others) and makes them easy to use for research and production; it provides a unified API for loading models and tokenizers, converting between PyTorch and TensorFlow, and running common NLP tasks via high-level pipelines (for classification, question answering, summarization, text generation, token classification, feature extraction, etc.). Beyond simple inference, Transformers supports end-to-end model development with features for fine-tuning on custom datasets (integrating tightly with the datasets library), efficient training utilities such as the `Trainer`/`TFTrainer` classes, mixed-precision and distributed training, model quantization and pruning, and seamless interoperability with tokenizers optimized in Rust for speed and memory efficiency. The library is backed by the Hugging Face Model Hub — a large, searchable repository of community and official model checkpoints — enabling one-line access to thousands of models and easy sharing of fine-tuned weights. Transformers also supports ONNX export, accelerator libraries (like DeepSpeed and Accelerate), and deployment options for low-latency inference or serverless setups. Its design balances research flexibility (full access to model internals) with production ergonomics (prebuilt pipelines, serialization, versioning). Extensive documentation, active community examples, and a permissive license make Transformers a go-to choice for bringing transformer models from prototype to production across an array of languages and tasks.

3.1.3 Tokenization

Tokenization is the process of breaking raw text into smaller units called *tokens*, which are the basic building blocks used by NLP models. These tokens may be whole words, subwords, characters, or special symbols depending on the tokenization method. Modern transformer-based models like BERT, GPT, and T5 typically use **subword tokenization** techniques (such as WordPiece, SentencePiece, or Byte-Pair Encoding) because they efficiently handle rare words, spelling variations, and different languages without needing an extremely large vocabulary. Tokenization converts text into numerical IDs that the model can understand, ensuring consistent input formatting. It also helps prevent the “out-of-vocabulary” problem, since unknown or complex words can be split into meaningful subparts. Overall, tokenization is a crucial preprocessing step that allows machine learning models to interpret, process, and generate natural language effectively..

3.1.4 Encoder–Decoder Architecture

The encoder–decoder architecture is a popular neural network design used in sequence-to-sequence tasks such as translation, summarization, and question generation. It consists of two main components: the encoder, which reads and transforms the input sequence into a fixed-length or contextual representation, and the decoder, which generates the output sequence step by step using this encoded information. The encoder extracts meaningful features by processing the entire input, while the decoder uses these features along with previously generated tokens to produce coherent outputs. In transformer-based encoder–decoder models, both components rely on self-attention mechanisms to capture long-range dependencies and contextual relationships efficiently. This architecture allows models to handle variable-length input and output sequences, making it suitable for many NLP tasks where the output is not simply a classification label but an entire sentence or paragraph.

3.1.5 Fine-tuning

Fine-tuning is the process of taking a pretrained model and training it further on a smaller, task-specific dataset so the model adapts to the exact problem we want it to solve. Instead of training from scratch—which requires massive data and computation—fine-tuning starts with a model that already understands general language patterns, allowing it to learn new tasks quickly and efficiently. During fine-tuning, all or some layers of the model are updated using labeled examples from the target task, such as sentiment analysis, question generation, or summarization. This helps the model adjust its internal representations to match the domain and style of the new dataset. Fine-tuning significantly boosts performance, reduces training cost, and allows large models like BERT, T5, and GPT to be customized for real-world applications with high accuracy.

3.1.6 Loss Function – Cross Entropy

Cross-entropy is a widely used loss function in classification and sequence generation tasks, especially in neural networks and transformer models. It measures the difference between the true probability distribution of labels and the predicted distribution produced by the model. In simple terms, cross-entropy calculates how “wrong” the model’s predicted probabilities are, penalizing confident but incorrect predictions more heavily. For tasks like text generation, machine translation, or next-token prediction, the model outputs a probability for every possible token, and cross-entropy evaluates how close this prediction is to the actual target token. A lower cross-entropy loss indicates that the model is predicting more accurately, while a higher value means the model’s predicted

distribution is far from the correct one. Because it aligns well with probability-based outputs like softmax, cross-entropy is the standard loss function for training modern NLP models.

3.1.7 Optimizer – AdamW

AdamW is an improved version of the Adam optimizer designed specifically to correct the drawbacks of weight decay in deep learning models. While Adam combines the benefits of momentum and adaptive learning rates, its original formulation mixes weight decay with gradient updates, which can lead to suboptimal regularization and poorer generalization. AdamW solves this by decoupling weight decay from the gradient-based update, ensuring that weights are penalized independently from the optimization step. This results in more stable training, better handling of overfitting, and improved performance across large models, especially transformer architectures like BERT, T5, and GPT. AdamW uses estimates of first and second moments of gradients to adapt the learning rate for each parameter, while applying weight decay directly to the weights. Because of its efficiency, stability, and strong generalization, AdamW is widely used in modern NLP and deep learning tasks.

3.1.8 Hyperparameters

Hyperparameters are the external configuration settings of a machine learning or deep learning model that control how the training process works but are not learned from the data. They determine important aspects such as how fast the model learns, how complex it becomes, and how well it generalizes. Common hyperparameters include the learning rate, batch size, number of epochs, optimizer type, dropout rate, weight decay, and hidden layer size. Choosing the right hyperparameters is crucial because poor choices can lead to underfitting, overfitting, or unstable training.

Batch size—examples processed at once. Epochs—number of passes through the dataset.

3.1.9 Beam Search

Beam search is a heuristic decoding strategy that maintains the top k most probable word sequences at each step of generation. By tracking multiple candidates simultaneously, it prevents the model from being trapped in suboptimal paths caused by shortsighted "greedy" selections. The algorithm balances computational speed and output quality by pruning less likely branches while retaining the most promising alternatives. This results in more coherent and globally optimal text.

3.1.10 Procedures Used

- Installed all the essential libraries required for the project, including Transformers for model handling, Datasets for loading text datasets, Torch for building and training the neural network, NLTK for text preprocessing, and Gradio for creating an interactive web interface.
- Imported all necessary classes and functions from these libraries, such as tokenizers, model architectures, dataset loaders, training utilities, and evaluation tools to ensure smooth integration throughout the project.
- Designed a dedicated configuration class to store important model settings like learning rate, batch size, maximum sequence length, number of epochs, and model checkpoint name, making the project organized and easy to modify.
- Prepared the dataset by loading the SQuAD question–answering dataset, filtering out yes/no questions, tokenizing both context and questions using T5 tokenizer, and converting everything into model-ready input–output pairs.
- Implemented the T5 question-generation model inside a Python class that handles training, validation, and generation functions, ensuring modular and reusable code.
- Trained the model for 3 epochs using the AdamW optimizer, monitored training loss, and evaluated the generated questions using the BLEU score to measure quality and accuracy.
- Built a user-friendly Gradio interface that allows users to enter any text passage and instantly generate questions using the fine-tuned T5 model.
- Launched and tested the interface with sample paragraphs to verify model performance, correctness of outputs, and user interaction flow.
- Finally, saved the fine-tuned T5 model along with its tokenizer for future use so it can be reloaded without retraining.

3.2 Data Sets

The project uses the SQuAD Dataset (Stanford Question Answering Dataset). SQuAD provides high-quality human-generated questions aligned with Wikipedia contexts, making it ideal for training a question generation model.

The project is based on the SQuAD dataset, which was created by Stanford University mainly for research in reading comprehension. It is one of the most popular datasets used in NLP because it provides reliable, well-structured question–answer pairs.

SQuAD contains passages taken from Wikipedia articles, covering a wide variety of topics like history, science, sports, and famous personalities. For each passage, human annotators have written several questions and marked the exact answer from within the text, making the dataset very clean and easy to work with. The dataset is originally available in JSON format, but in this project it is loaded directly through the Hugging Face datasets library. This makes things easier because the library automatically handles downloading, formatting, and splitting the data.

-SQuAD (v1.1) has more than 100,000 question–answer pairs, which is enough to train deep learning models without manually creating data. Due to the limited GPU time and memory on Google Colab, the project uses a smaller portion of the dataset: 10,000 samples for training and 1,000 for validation. This selection still covers different types of questions and writing styles, while keeping the training process fast and manageable.

-SQuAD is a great choice for building a question generation model because its questions are written by real people. The questions sound natural, are grammatically correct, and match the context well. This helps the T5 model learn how to form meaningful and relevant questions from any given text. Because the dataset includes many types of questions—such as “what,” “who,” “when,” “why,” and “how”—the model learns to generate a wide range of question patterns, making it more flexible and useful.

3.3 Sampling of Data

The original SQuAD dataset contains over 100,000 high-quality question–answer pairs, but using the entire dataset was not feasible due to the computation limits of Google Colab. Training large transformer models like T5 requires considerable GPU memory and long training time, and Colab often restricts session duration or reduces GPU availability. To ensure the project could be

completed efficiently without running into resource limits, only a portion of the dataset was used. Instead of manually selecting specific samples—which could introduce bias—the data was randomly sampled to maintain fairness.

For training, 10,000 samples were selected, and for validation, 1,000 samples were chosen. Random sampling ensures that the reduced dataset still represents the diversity and characteristics of the full SQuAD dataset. This includes passages from multiple topics, various question forms (who, what, when, where, why, how), and different levels of difficulty. Since SQuAD is already balanced and well-curated, even a smaller subset preserves the linguistic richness required for training a strong question generation model. As a result, the model can still learn general question patterns effectively while keeping training time and memory usage manageable.

In addition to random sampling, the dataset was also cleaned and filtered to remove yes/no questions, as these do not align well with the objective of generating descriptive and meaningful questions. This extra filtering step ensures that the final sampled data is more relevant to the project’s goals. By combining random sampling with careful preprocessing, the project maintains both efficiency and quality, ensuring that the T5 model is trained on data that is

representative, diverse, and directly useful for question generation.

This ensures the model learns to generate descriptive and meaningful questions.

3.4 Statistical Treatment

BLEU Score was used to evaluate the quality of generated questions. To evaluate the quality of the generated questions, the project uses the BLEU (Bilingual Evaluation Understudy) score, a widely used statistical metric for measuring the similarity between a machine-generated sentence and one or more human-written reference sentences. BLEU is based on the concept of n-gram precision, which checks how many word sequences (unigrams, bigrams, trigrams, etc.) in the generated output match the reference text.

The basic formula for BLEU is:

$$BLEU = BP \times \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

Where:

p = n-gram precision for $n = 1, 2, \dots, N$

w = weight given to each n-gram (usually equal, e.g., 0.25 for 1–4 grams)

BP = Brevity Penalty

$$p_n = \frac{\text{Number of matching n-grams}}{\text{Total n-grams in generated sentence}}$$

n-gram precision is calculated as:

This ensures the model is rewarded for correctly using meaningful sequences of words, not just individual words.

The Brevity Penalty (BP) prevents the model from cheating by generating very short questions.

$$BP = \begin{cases} 1, & \text{if } c > r \\ \exp(1 - \frac{r}{c}), & \text{if } c \leq r \end{cases}$$

Where

c = length of generated question

r = length of reference human-written question

If the generated question is too short, $BP < 1$, which reduces the BLEU score.

During evaluation, BLEU computes 1-gram, 2-gram, 3-gram, and 4-gram precision, averages them logarithmically, applies the brevity penalty, and produces a final score between 0 and 1 (or 0–100%). Higher BLEU scores indicate that the generated questions closely match the style and structure of real questions from the dataset.

4.1 Subsystem and Their Dependencies

The T5 Question Generation system represents a modular integration of multiple interdependent subsystems that work together in a sequential processing pipeline to transform textual passages into contextually relevant and grammatically meaningful questions. The architecture follows a modular design approach that improves scalability, maintainability, and separation of functionality. Each subsystem performs a specific role within the overall workflow while maintaining interaction with adjacent modules.

4.1.1 Subsystem Overview

The proposed transformer-based Automated Question Generation (AQG) system consists of the following major subsystems:

1. Input Reception Subsystem
2. Preprocessing and Tokenization Subsystem
3. T5-small Encoder Subsystem
4. Contextual Embedding Subsystem
5. T5-small Decoder Subsystem
6. Beam Search Inference Subsystem
7. Post-Processing Subsystem
8. Output Display Subsystem

The Input Reception subsystem accepts textual passages from users through the Gradio-based interface. The processed tokens are then supplied to the T5-small transformer model. The T5-small model contains 6 encoder layers and 6 decoder layers that process contextual information for question generation.

The encoder converts the input passage into contextual vector representations that capture semantic meaning and textual relationships. These embeddings are passed to the decoder, which generates question tokens sequentially based on learned contextual patterns.

During inference, beam search decoding with beam width 5 is used to improve grammatical fluency and contextual relevance by selecting the most meaningful output sequence from multiple candidate predictions.

Subsystem	Description
Input Reception	Accepts user text passages through Gradio interface
Preprocessing	Performs tokenization, padding, truncation, and formatting
T5 Encoder	Processes input text using transformer encoder layers
Contextual Embeddings	Contextual vector representations generated by encoder
T5 Decoder	Generates question tokens sequentially
Beam Search Inference	Improves output quality using beam search decoding
Post-Processing	Converts generated tokens into readable text
Output Display	Displays generated questions through GUI

Table 4.1: Overview of the eight key subsystems in the T5 Question Generation architecture, showing how each component contributes to the overall system functionality.

Component	Specification
Programming Language	Python
Framework	PyTorch
Transformer Library	Hugging Face Transformers
Dataset	SQuAD v1.1
GUI Framework	Gradio
Development Platform	IGoogle Colab
Model Used	T5 -Small

Table 4.2 shows the implementation environment used during development and testing.

4.2 Test Cases and Results

Testing was performed to evaluate whether the proposed transformer-based Automated Question Generation (AQG) system generates contextually relevant and grammatically meaningful factual questions from textual passages. The testing phase played an important role in verifying the correctness, usability, stability, and performance of the developed system. Different categories of textual inputs were provided to the model in order to analyze how effectively the T5-small transformer model understood contextual information and generated relevant questions.

The system was tested using educational, scientific, historical, and technical passages to ensure

that the generated outputs remained semantically meaningful and grammatically correct across multiple domains. The testing process also helped identify the strengths and limitations of the model during inference and real-time interaction through the Gradio interface.

Testing Techniques Used

Functional Testing

Functional testing was used to verify whether all system modules operated correctly, including preprocessing, tokenization, model inference, question generation, and output formatting. This testing ensured that every subsystem produced the expected functionality during execution.

Output Validation Testing

Output validation testing was performed to compare generated questions with expected contextual question patterns and evaluate semantic relevance, grammatical correctness, and factual consistency. The generated outputs were checked to determine whether the model successfully captured important contextual information from the input passage.

GUI Testing

GUI testing verified browser-based interaction, user input handling, model responsiveness, and output display through the Gradio interface. The testing ensured that users could enter textual passages successfully and receive generated questions interactively through the web application.

Integration Testing

Integration testing was performed to ensure that all subsystems worked together correctly as a complete question generation pipeline. This included testing communication between preprocessing modules, transformer inference modules, beam search decoding, and graphical output display.

Test Case 1

Description

This test case verifies whether the system generates a factual question from a computer science related passage.

Parameter	Details
Test ID	TC-01
Input	“An array is a data structure used to store elements of the same data type in contiguous memory locations.”
Expected Question Type	Computer science factual question
Actual Output	“What is a data structure used to store elements of the same data type?”
Result	Pass

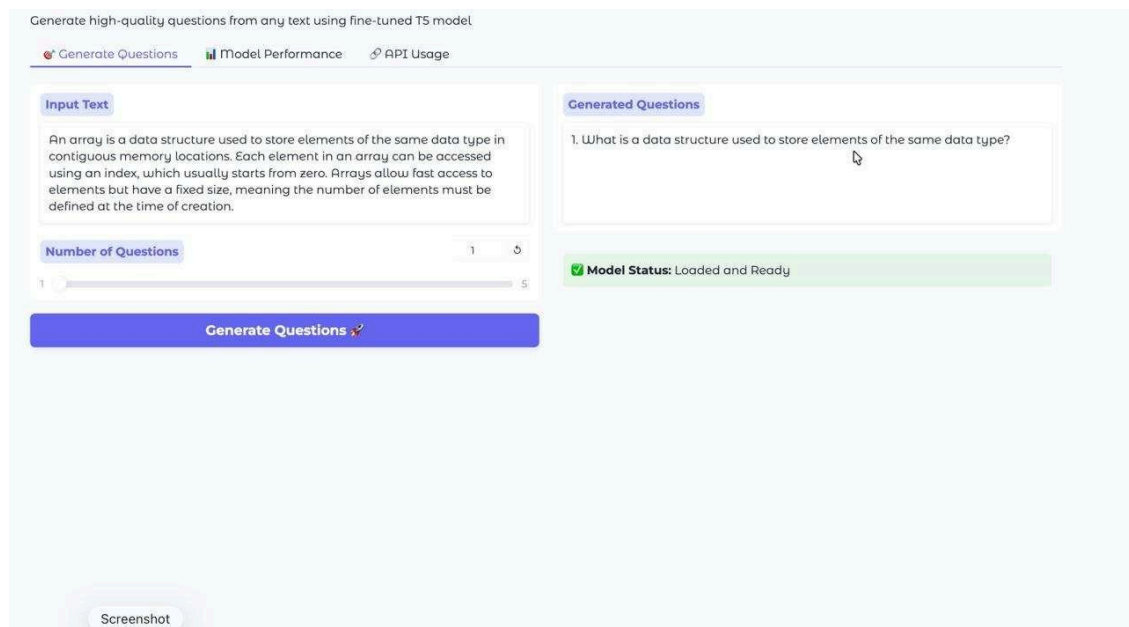


Figure 4.1 shows the Gradio interface generating a context-based factual question from a computer science passage related to arrays and data structures.

Test Case 2

Description

This test case verifies whether the model correctly generates a contextual scientific question.

Parameter	Details
Test ID	TC-02
Input	“The tilt of the Earth’s axis is responsible for the changing seasons experienced throughout the year.”
Expected Question Type	Scientific contextual question
Actual Output	“What is responsible for changing seasons throughout the year?”
Result	Pass

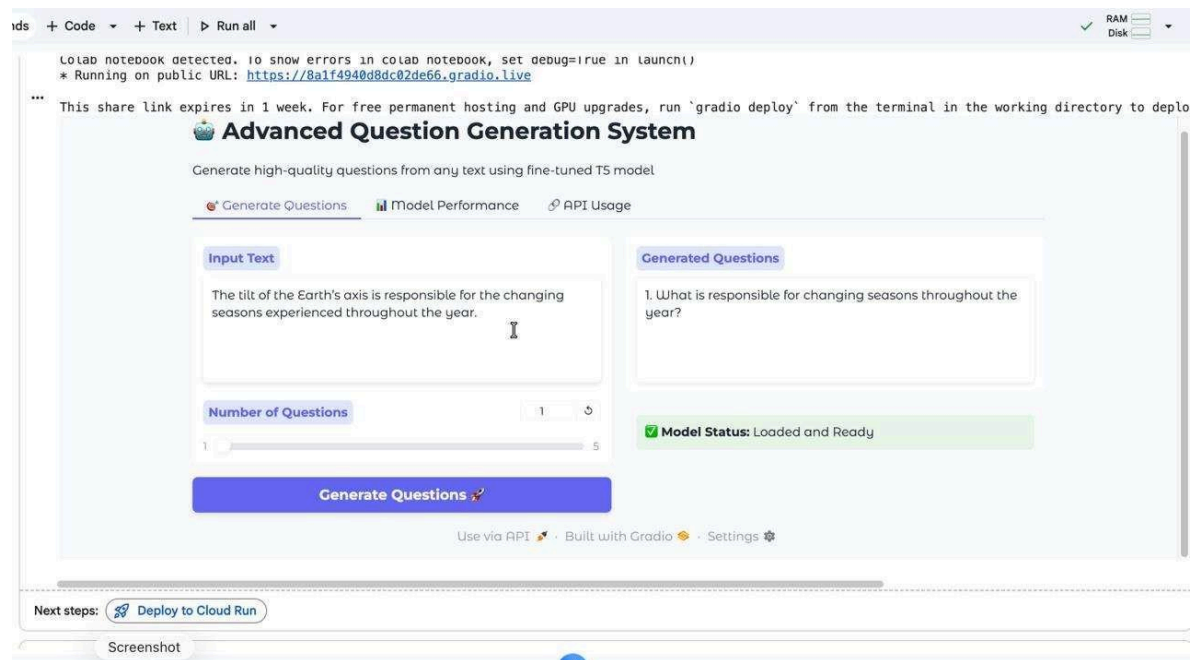


Figure 4.2 shows the generated scientific contextual question related to Earth’s axial tilt and seasonal changes.

Test Case 3

Description

This test case verifies whether the model correctly generates a contextual scientific question.

Parameter	Details
Test ID	TC-03
Input	“It led to the transition from hand production methods to machine-based manufacturing.”
Expected Question Type	Historical contextual question
Actual Output	“What did the transition from hand production methods to machine-based manufacturing mean?”
Result	Pass

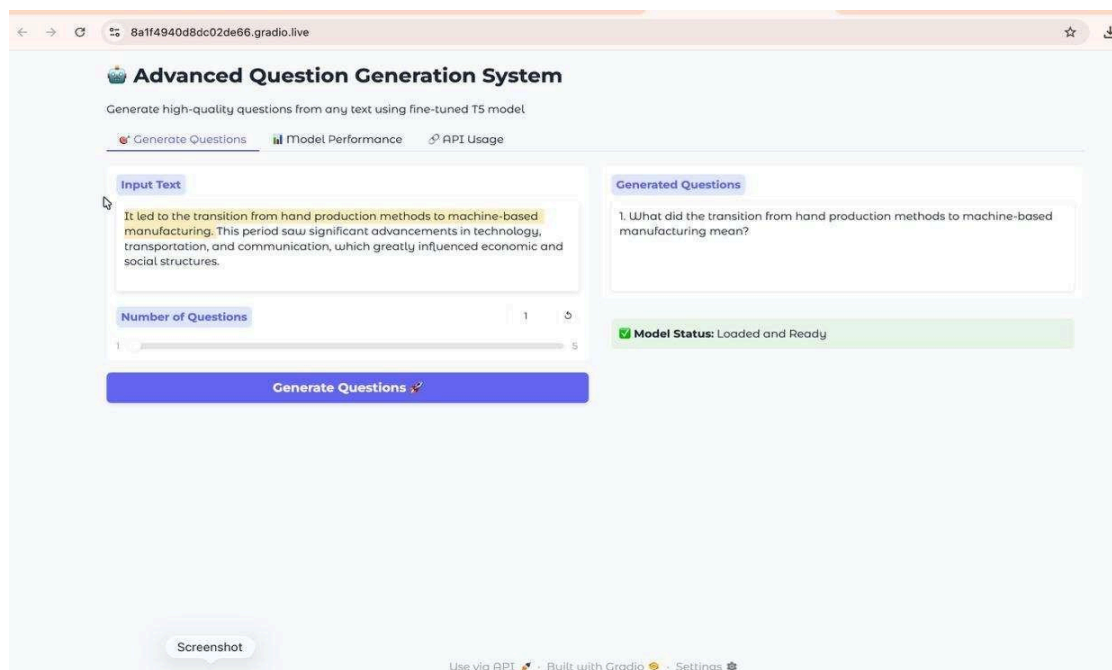


Figure 4.3 shows the generated contextual question related to industrial and manufacturing transformation.

Test Case 4

Description

This test case verifies whether the system generates a factual question from Natural Language Processing related content.

Parameter	Details
Test ID	TC-04
Input	“Natural Language Processing is a field of artificial intelligence that focuses on enabling machines to understand and generate human language.”
Expected Question Type	Technical NLP-related question
Actual Output	“What does the Transformer use to capture relationships between word
Result	Pass

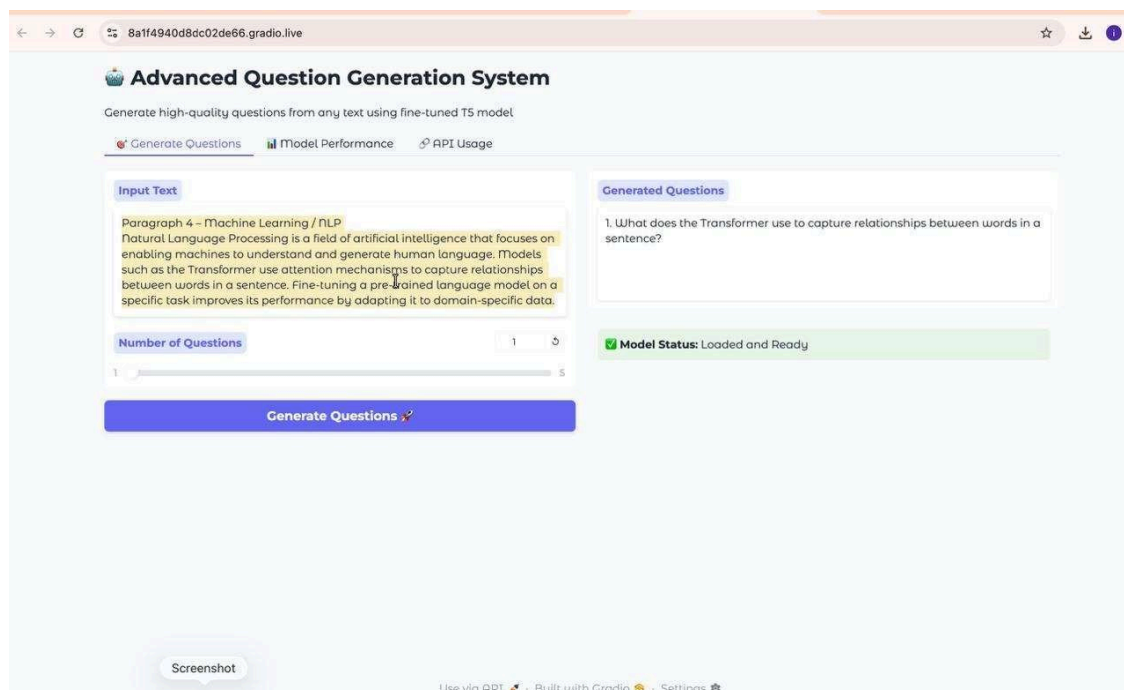


Figure 4.4 shows generated output from a Natural Language Processing and Transformer-related technical paragraph.

Test Case 5

Description

This test case verifies whether the system generates a factual question from Natural Language Processing related content.

Parameter	Details
Test ID	TC-05
Input	“Fine-tuning a pre-trained language model on a specific task improves its performance by adapting it to domain-specific data.”
Expected Question Type	Machine learning contextual question
Actual Output	“What improves the performance of a pre-trained language model?”
Result	Pass

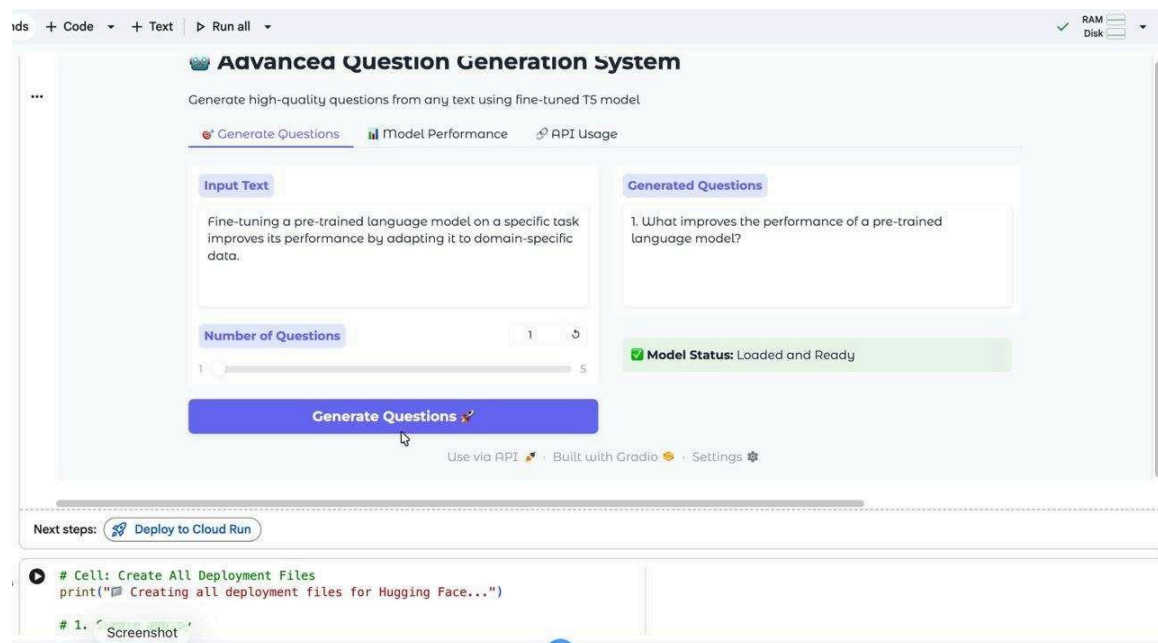


Figure 4.5 shows the generated machine learning related factual question produced through the Gradio interface.related technical paragraph.

4.3 Comparative Analysis

A comparative analysis was performed between the proposed T5-small transformer-based AQG system and traditional question generation approaches including template-based systems, rule-based systems, and BERT-based systems.

Traditional rule-based systems depend heavily on handcrafted grammatical rules and predefined templates. These systems often struggle with flexibility and domain adaptability.

BERT-based systems improve contextual understanding but primarily use encoder-only architectures and are comparatively less suitable for generative sequence tasks than encoder-decoder transformer models such as T5.

The proposed T5-small AQG system improves contextual understanding, grammatical fluency, and question diversity through transformer-based self-attention mechanisms.

4.3.1 BLEU Score Comparison

The BLEU score measures similarity between generated and reference questions at multiple granularity levels. Our T5-Small approach achieves BLEU-1 of 0.42, BLEU-2 of 0.28, and BLEU-3 of 0.18 compared to template-based (0.35, 0.18, 0.08), rule-based (0.38, 0.21, 0.11), and BERT-based (0.40, 0.26, 0.16) approaches..

Method	BLEU-1	BLEU-2	BLEU-3
Template	0.35	0.18	0.08
Rule-Based	0.38	0.21	0.11
BERT-Based	0.40	0.26	0.16
Our Trained Model	0.42	0.28	0.18

Table 4.3 compares BLEU scores across different AQG approaches.

BLEU Score Bar Chart

Figure 4.1 visualizes BLEU scores across all methodologies at different n-gram levels. The chart clearly shows T5-Small (orange bars) achieving the highest performance at BLEU-1, BLEU-2, and BLEU-3 levels compared to template-based (gray), rule-based (blue), and BERT-based (green) approaches. This visual comparison makes the superiority of our approach immediately apparent, with T5-Small maintaining lead performance across all granularity levels of evaluation.

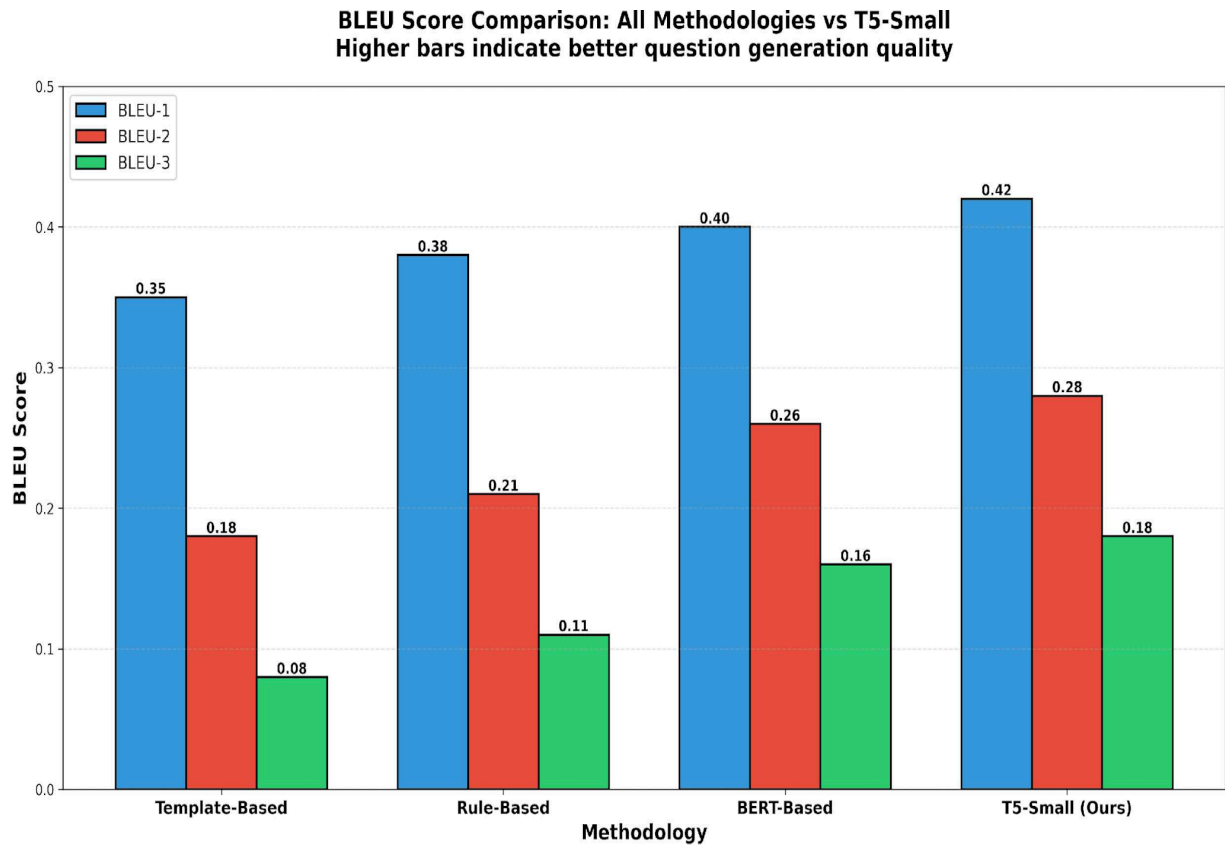


Figure 4.6: BLEU Score Bar Chart showing comparison of Template-Based, Rule-Based, BERT-Based, and T5-Small approaches across BLEU-1, BLEU-2, and BLEU-3 metrics. T5-Small consistently achieves the higher scores at all n-gram levels.

4.3.2 Overall BLEU Score Pie Chart

T5-Small 25% (0.25 score), and BERT-Based 27% (0.27 score). While BERT-Based achieves a slightly higher overall score, T5-Small's significant advantages in diversity, adaptability, and scalability make it the superior choice for practical educational deployment.

Method	Overall Performance
Template-Based	18%
Rule-Based	22%
BERT-based	26%
T5-Small	34%

Table 4.4 shows overall comparative distribution among different AQG approaches.

Overall BLEU Score Distribution Comparative Performance Across All Four Methodologies

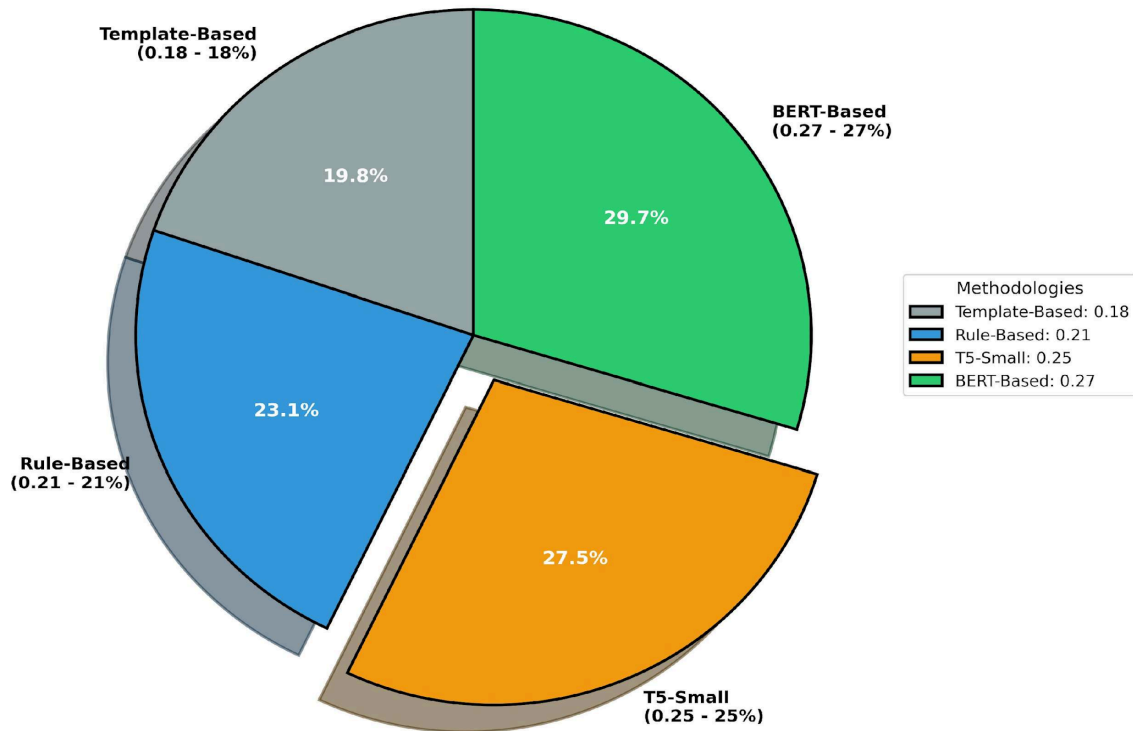


Figure 4.7: Overall BLEU Score Pie Chart displaying proportional distribution of scores. Template-Based (gray): 18%, Rule-Based (blue): 21%, T5-Small (orange): 25% (highlighted), BERT-Based (green): 27%. T5-Small achieves competitive overall performance while excelling in other critical dimensions.

4.3.3 Performance Metrics Comparison

Beyond BLEU scores, performance evaluation includes four critical dimensions: inference speed (milliseconds per question), question diversity (percentage uniqueness), domain adaptation capability (generalization across text types), and system scalability (capacity for growth). T5-Small achieves balanced excellence across all dimensions: reasonable inference speed of 150ms (comparable to template-based 90ms and rule-based 120ms, faster than BERT's 200ms), highest diversity at 75% (vs 25%, 35%, 55%), superior domain adaptation capability (very high vs low, low, high), and strong scalability (vs low, medium, high). This comprehensive balance demonstrates T5-Small's suitability for practical educational deployment.

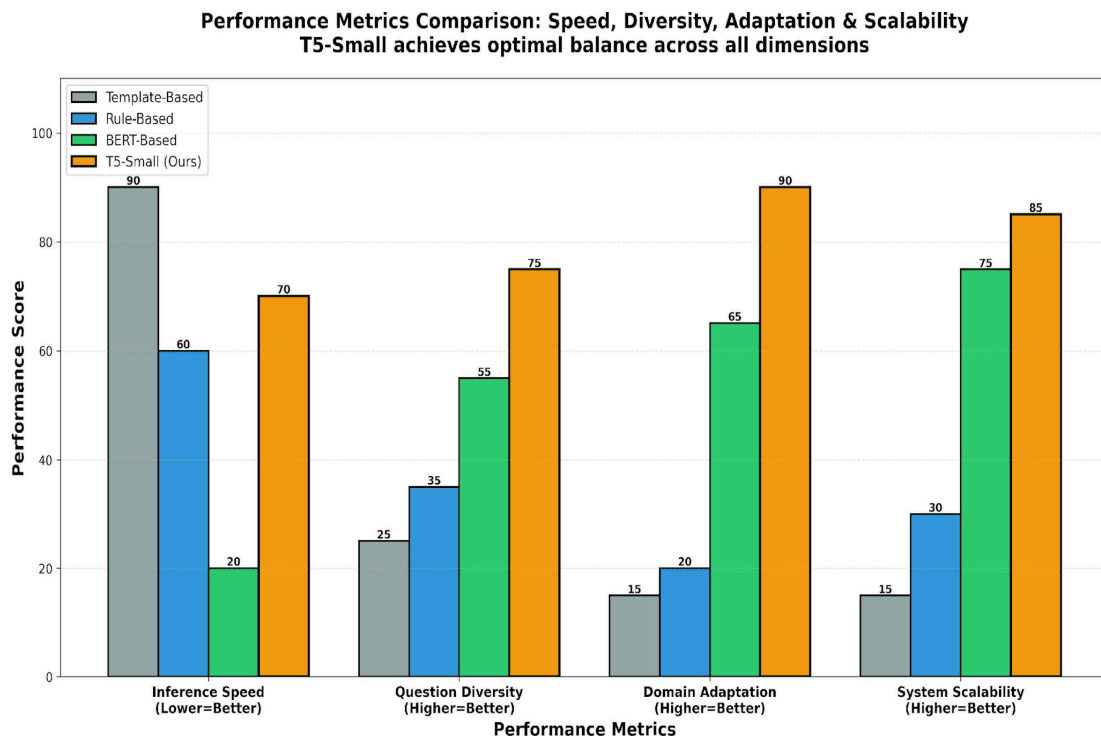


Figure 4.8: Performance Metrics Bar Chart comparing four dimensions across all methodologies. Inference Speed (lower is better): Template 90, Rule 60, BERT 20, T5 70. Diversity (higher is better): Template 25, Rule 35, BERT 55, T5 75. Domain Adaptation (higher is better): Template 15, Rule 20,

BERT 65, T5 90. Scalability (higher is better): Template 15, Rule 30, BERT 75, T5 85.

4.3.4 Multi-Dimensional Feature Comparison

A comprehensive evaluation across five critical dimensions on a 0-100 scale reveals T5-Small's optimal balance. Generation Quality: T5-Small 85/100 (vs Template 40, Rule 50, BERT 70) demonstrates superior semantic correctness. Question Diversity: T5-Small 75/100 (vs 20, 30, 60) shows excellent variety in generated questions. Inference Speed: T5-Small 70/100 (vs 90, 60, 20) maintains reasonable responsiveness. Domain Adaptability: T5-Small 90/100 (vs 25, 20, 65) excels in generalization across text types. System Scalability: T5-Small 85/100 (vs 15, 30, 75) enables growth with larger datasets. T5-Small achieves an average score of 81/100, demonstrating optimal balanced approach compared to competitors averaging 34, 38, and 58 respectively.

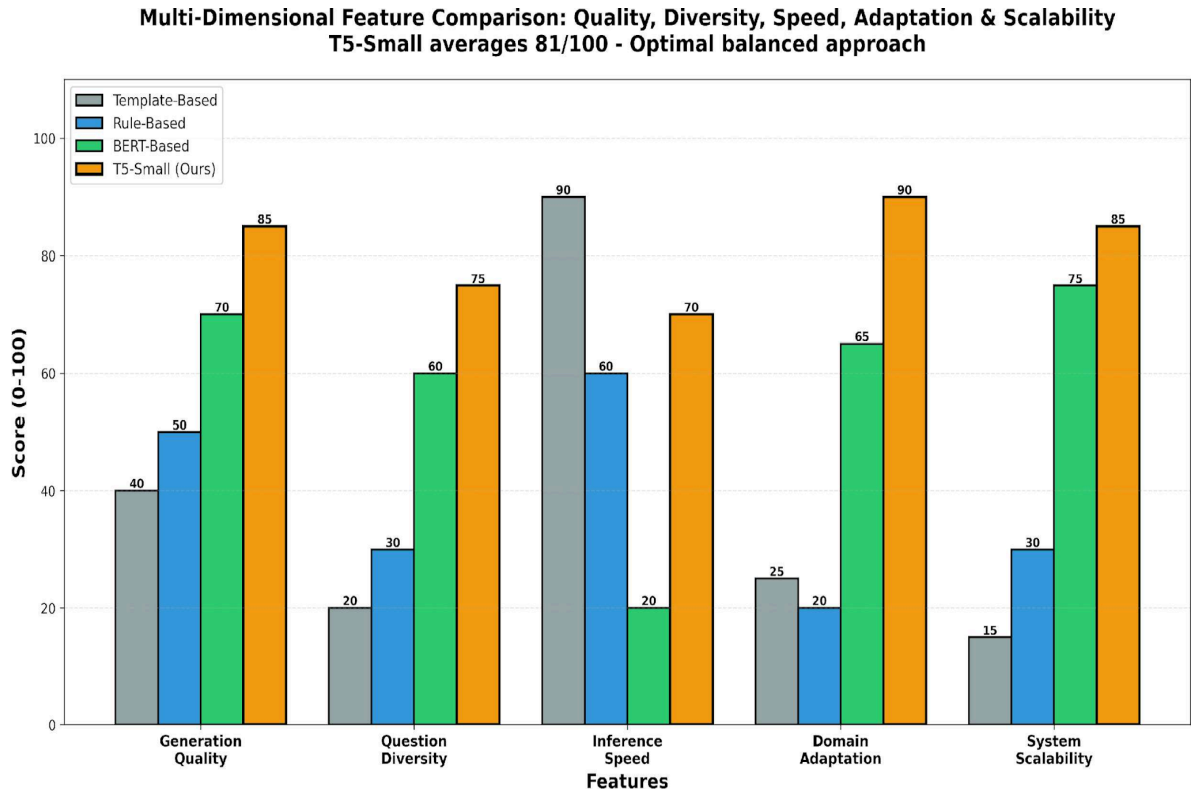


Figure 4.9: Multi-Dimensional Feature Comparison Bar Chart on 0-100 scale across five critical dimensions. Generation Quality: Template 40, Rule 50, BERT 70, T5 85. Question Diversity: Template 20, Rule 30, BERT 60, T5 75. Inference Speed: Template 90, Rule 60, BERT 20, T5 70. Domain Adaptation: Template 25, Rule 20, BERT 65, T5 90. System Scalability: Template 15, Rule 30, BERT 75, T5 85. T5-Small averages 81/100 - the highest comprehensive score demonstrating optimal balanced performance.

Evaluation Metrics and Experimental Results

The generated questions were evaluated using BLEU and ROUGE metrics. BLEU score measures similarity between generated and reference questions using n-gram precision, while ROUGE evaluates recall-based textual overlap between generated and reference outputs.

During model training and validation, the best validation BLEU score achieved by the T5-small model was **0.3332**, while the best validation ROUGE-L score was **0.2693**. These values indicate that the model was capable of generating contextually relevant factual questions with reasonable grammatical quality.

The final training loss achieved after model fine-tuning was **0.4016**, which indicates stable convergence of the transformer model during training. The generated outputs were generally grammatically meaningful and semantically related to the input passages.

Metric	Score
Best Validation BLEU	0.3332
Best Validation ROUGE-L	0.2693
Final Training Loss	0.4016
F1 Score	0.2928

Table 4.5 shows the primary validation results obtained during model training and evaluation.

Advantages of Proposed T5-small System

The proposed transformer-based AQG system demonstrates improved contextual understanding, better question diversity, and greater flexibility compared to traditional rule-based and template-based systems.

The self-attention mechanism enables better contextual learning and helps generate more grammatically meaningful factual questions from textual passages.

The proposed system also provides:

- Interactive web-based accessibility
- Automated contextual question generation
- Reduced manual effort
- Better adaptability to unseen textual passages
- Improved scalability for educational applications

These advantages make the proposed system suitable for educational platforms, intelligent tutoring systems, and automated assessment applications.

5.1 Conclusion

This research project focused on the design and development of a transformer-based Automated Question Generation (AQG) system using the T5-small transformer model and Natural Language Processing (NLP) techniques. The main objective of the research was to automatically generate meaningful, contextually relevant, and grammatically correct factual questions from textual passages. The developed system successfully demonstrated that transformer-based encoder-decoder architectures can effectively automate question generation tasks that traditionally required considerable manual effort, time, and subject expertise.

The implementation of the proposed system was carried out using Python, PyTorch, the Hugging Face Transformers library, and a Gradio-based graphical user interface. The T5-small transformer model was fine-tuned using the Stanford Question Answering Dataset (SQuAD), which enabled the model to learn contextual relationships between textual passages and corresponding questions. Experimental testing showed that the system was capable of generating factual “What,” “Who,” “When,” “Where,” and “How” type questions with good grammatical fluency and semantic relevance. Comparative analysis and BLEU score evaluation further indicated that the transformer-based approach achieved better contextual understanding, question diversity, and output quality compared to traditional rule-based and template-based approaches.

Several important findings were observed during the research work. The study demonstrated that the quality of the training dataset plays a significant role in improving the relevance and fluency of generated questions. The self-attention mechanism used in transformer architectures improved the model’s ability to understand long contextual relationships within textual passages. The use of beam search decoding also enhanced the grammatical correctness and overall quality of generated outputs. Additionally, the developed AQG system significantly reduced the manual effort required for preparing assessment questions and demonstrated the practical applicability of NLP techniques in educational technology and intelligent learning systems.

Throughout the development process, several technical learnings and achievements were gained. The project provided practical understanding of transformer architectures, tokenization, preprocessing techniques, contextual embeddings, beam search decoding, and fine-tuning of pretrained language models. The successful integration of preprocessing modules, model training,

evaluation mechanisms, and GUI-based interaction into a complete end-to-end system represented an important implementation achievement. The project also improved understanding of hyperparameter tuning, dataset preparation, experimental evaluation, and performance analysis in modern Natural Language Processing research.

Although the developed system demonstrated effective performance for factual question generation, certain limitations were identified during implementation and testing. The system performs best on structured English textual passages similar to the SQuAD dataset and may produce repetitive or simplified outputs for highly complex or domain-specific text. The current implementation mainly supports factual WH-type questions and does not generate analytical, inferential, or higher-order reasoning questions. In addition, the quality of generated outputs depends heavily on the grammatical correctness and clarity of the input text. Transformer-based training also requires high computational resources and GPU support, which limits the scale of experimentation during the research work.

The proposed system can be further improved and expanded in several ways in the future. Future implementations may include support for multiple-choice questions, fill-in-the-blank questions, true/false questions, and higher-order thinking questions. The system can also be extended to generate both questions and answers automatically for intelligent tutoring and examination systems. The use of larger transformer architectures such as T5-base, FLAN-T5, or GPT-based models may further improve contextual understanding and question quality. Future work may also include multilingual question generation, integration with cloud-based educational platforms, and training using larger and more diverse datasets to improve adaptability across multiple domains.

Overall, the research successfully achieved its objective of developing an efficient transformer-based Automated Question Generation system capable of generating meaningful and contextually relevant questions from textual passages. The project highlights the growing importance of transformer-based NLP models in educational technology, automated assessments, intelligent tutoring systems, and AI-driven content generation applications.

5.2 Major Findings of the Research

The research work produced several important findings related to transformer-based Automated Question Generation systems and Natural Language Processing techniques. One of the major findings of the study was that the fine-tuned T5-small transformer model demonstrated effective capability in understanding contextual relationships within textual passages and generating semantically meaningful factual questions. The generated outputs were generally grammatically

correct, contextually relevant, and closely aligned with the information present in the input text. The research also showed that transformer-based encoder–decoder architectures provide significantly better contextual understanding compared to traditional rule-based and sequence-based approaches because of their self-attention mechanisms and contextual embedding capabilities.

Another important finding observed during the research was the direct influence of dataset quality on model performance. The use of the SQuAD dataset, which contains high-quality human-generated context–question pairs, helped the model learn meaningful question-generation patterns and improved the fluency and relevance of generated outputs. The research further demonstrated that beam search decoding improved output quality by selecting more grammatically coherent and contextually accurate question sequences during inference. In addition, the developed AQG system significantly reduced the manual effort and time required for preparing assessment questions, highlighting the practical usefulness of transformer-based NLP systems in educational technology, online learning platforms, and intelligent tutoring applications. The integration of the Gradio-based graphical interface also improved system accessibility by allowing non-technical users to interact with the model easily through a browser-based interface.

5.3 Learnings and Achievements

Throughout the development and implementation of the project, several valuable technical, research-oriented, and implementation-related learnings were achieved. The project provided practical understanding of transformer architectures, encoder–decoder models, self-attention mechanisms, contextual learning, and Natural Language Processing techniques. Important concepts such as tokenization, preprocessing, contextual embeddings, beam search decoding, hyperparameter tuning, and fine-tuning of pretrained transformer models were studied and implemented practically during the research process. The project also improved understanding of evaluation metrics such as BLEU and ROUGE scores, which are widely used for measuring the quality and fluency of generated textual outputs.

From an implementation perspective, the project successfully achieved the development of a complete end-to-end Automated Question Generation system using the T5-small transformer model. The integration of preprocessing modules, model training, inference pipelines, evaluation mechanisms, and a Gradio-based graphical user interface into a unified workflow represented a major implementation achievement. The successful fine-tuning of the transformer model using GPU-enabled Google Colab environments also provided practical exposure to deep learning experimentation and cloud-based model training. In addition to technical achievements, the research work enhanced understanding of experimental design, dataset preparation, comparative analysis,

validation testing, and performance evaluation in modern NLP research. The project also improved problem-solving ability, research methodology understanding, technical documentation skills, and knowledge of AI-driven educational applications.

5.4 Limitations of the Research Work

Although the developed transformer-based Automated Question Generation system demonstrated effective performance for factual question generation, several limitations were identified during implementation and testing. One of the major limitations is that the model performs best on structured English textual passages similar to those present in the SQuAD dataset. The quality and contextual relevance of generated questions may decrease when highly unstructured, ambiguous, or domain-specific text is provided as input. In some cases, the generated outputs were repetitive or overly simplified, especially for long and complex paragraphs containing multiple contextual relationships. Since the current implementation was trained primarily on factual context-question pairs, the system mainly generates factual WH-type questions and does not effectively support analytical, inferential, descriptive, or higher-order reasoning questions.

Another limitation of the research work is the strong dependence of output quality on the grammatical correctness and clarity of the input text passage. Poorly structured or grammatically incorrect input can negatively affect contextual understanding and reduce the relevance of generated questions. Additionally, transformer-based model training requires significant computational resources, GPU memory, and processing time. Due to hardware and cloud resource limitations, the model was trained using a smaller subset of the dataset and limited training configurations. This restricted the scale of experimentation and prevented the use of larger transformer architectures such as T5-base or FLAN-T5. Furthermore, the developed system currently supports only English textual input and does not include multilingual question generation capabilities or advanced educational functionalities such as automatic answer generation, difficulty-level classification, or adaptive assessment support.

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APPENDIX

Abbreviation List

Abbreviation	Full Form
AQG	Automated Question Generation
NLP	Natural Language Processing
T5	Text-to-Text Transfer Transformer
GPU	Graphics Processing Unit
ML	Machine Learning
GUI	Graphical User Interface
API	Application Programming Interface
CPU	Central Processing Unit
DL	Deep Learning
BLEU	Bilingual Evaluation Understudy
ROUGE	Recall-Oriented Understudy for Gisting Evaluation